EXP NO: 9

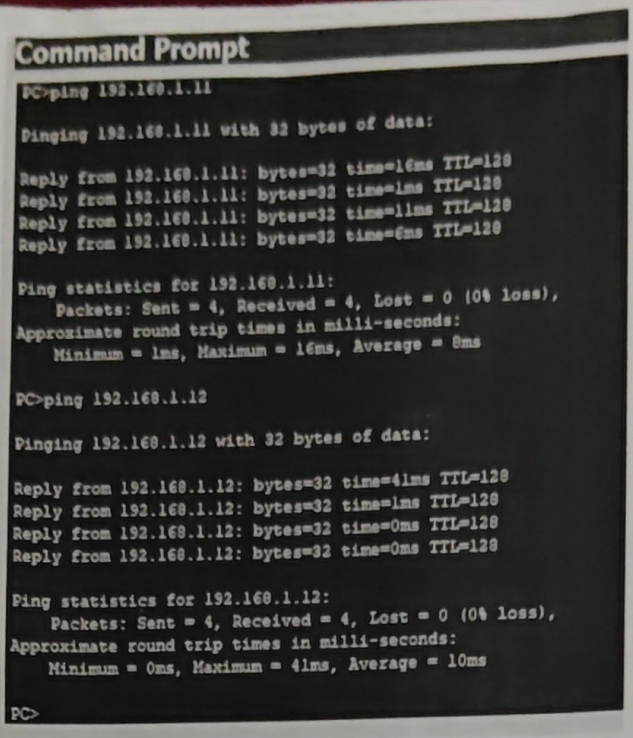
Date: 11/9/26 Implementation of subnetting in CISCO Packet Tracer simulator

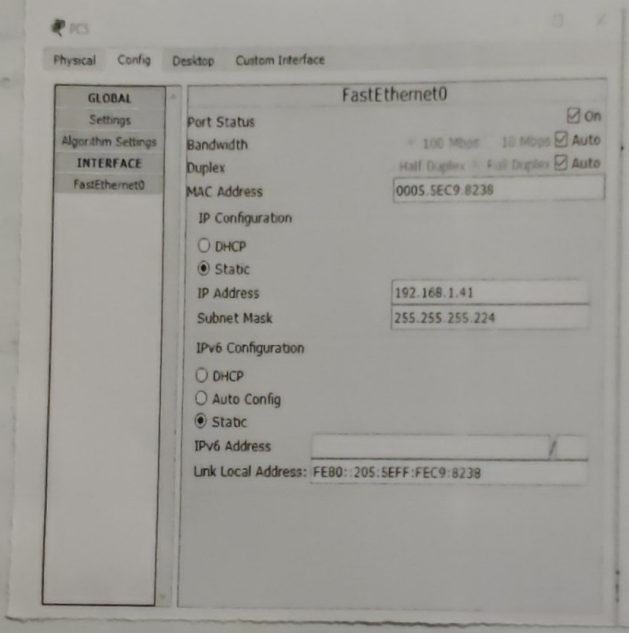
AIM:-

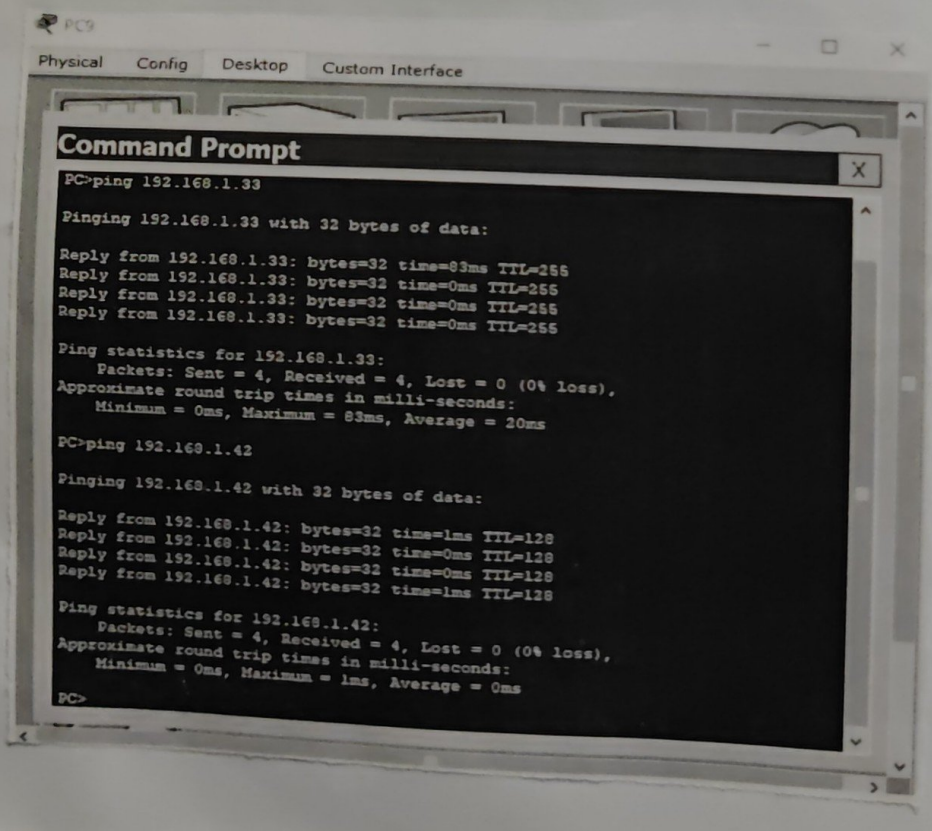
To implement subnetting in CISCO packet Tracer simulator.

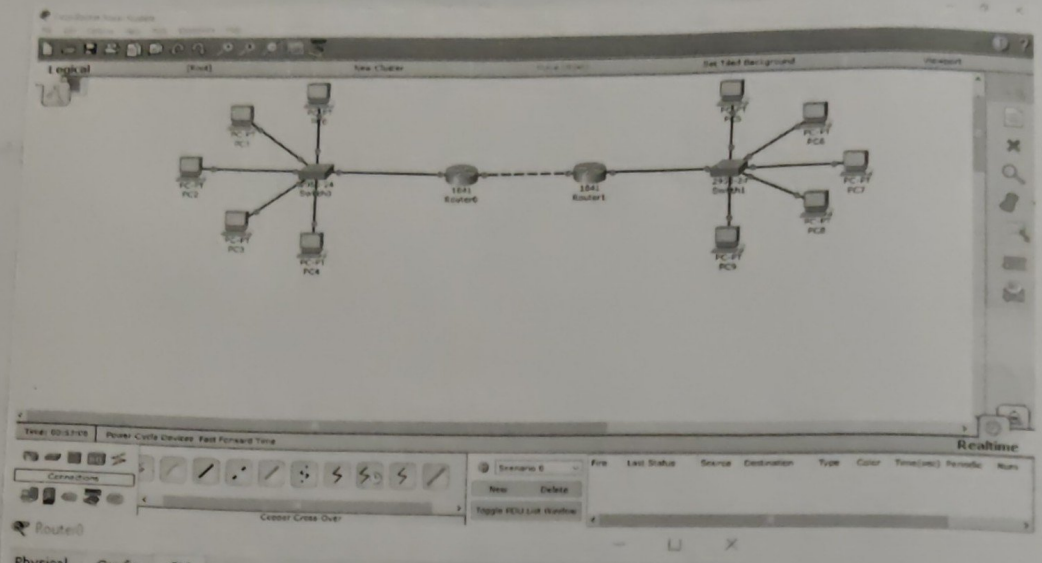
PROCEDURE:-

1. In Cisco Packet Tracer Create a new network topology
2. Add the required network devices:
   - ⇒ 2 routers (R0 and R1)
   - ⇒ 2 switches (S0 and S1)
   - ⇒ 10 PC's (PC0 - PC9)
3. Connect PC0-PC4 to switch S0 and PC5-PC9 to Switch S1 using suitable Ethernet cables.
4. Connect S0 to R0 and S1 to R1
5. Connect R0 and R1 using their available fast Ethernet interfaces.
6. Divide the network into /27 subnets. The subnet mask used is: 255.255.255.224.
7. Configure R0 using the CLI. Assign 192.168.1.1 to the interface connected to S0 and 192.168.1.65 to the interface connected to R1
8. Enable the router interfaces using the no shutdown commands.
9. Configure R1 using CLI. Assign 192.168.1.33 to the interface connected to S1 and 192.168.1.66 to the interface connected to R0
10. Enable the R1 interfaces using the no shutdown command in cmd.
11. Configure the IP addresses of PC0 - PC4 in the 1st subnet
    a) PC0 - 192.168.1.11
    b) PC1 - 192.168.1.12
    c) PC2 - 192.168.1.13


Command Prompt
PC>ping 192.168.1.11
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time=16ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=11ms TTL=128
Reply from 192.168.1.11: bytes=32 time=6ms TTL=128
Ping statistics for 192.168.1.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 16ms, Average = 8ms
PC>ping 192.168.1.12
Pinging 192.168.1.12 with 32 bytes of data:
Reply from 192.168.1.12: bytes=32 time=41ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=0ms TTL=128
Reply from 192.168.1.12: bytes=32 time=0ms TTL=128
Ping statistics for 192.168.1.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 41ms, Average = 10ms
PC>


PC5
Physical
Config
Desktop
Custom Interface
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0005.5EC9.8238
IP Configuration
DHCP
Static
IP Address
192.168.1.41
Subnet Mask
255.255.255.224
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address:
FE80::205:5EFF:FEC9:8238


PC9
Physical
Config
Desktop
Custom Interface
Command Prompt
PC>ping 192.168.1.33
Pinging 192.168.1.33 with 32 bytes of data:
Reply from 192.168.1.33: bytes=32 time=83ms TTL=256
Reply from 192.168.1.33: bytes=32 time=0ms TTL=256
Reply from 192.168.1.33: bytes=32 time=0ms TTL=256
Reply from 192.168.1.33: bytes=32 time=0ms TTL=256
Ping statistics for 192.168.1.33:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 83ms, Average = 20ms
PC>ping 192.168.1.42
Pinging 192.168.1.42 with 32 bytes of data:
Reply from 192.168.1.42: bytes=32 time=1ms TTL=128
Reply from 192.168.1.42: bytes=32 time=0ms TTL=128
Reply from 192.168.1.42: bytes=32 time=0ms TTL=128
Reply from 192.168.1.42: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.42:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
PC>



Physical Config CLI

IOS Command Line Interface

```
exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.1.1 255.255.255.224
% 192.168.1.0 overlaps with FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.1.65 255.255.255.224
% 192.168.1.64 overlaps with FastEthernet0/1
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#do show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.1.1     YES manual up                    down
FastEthernet0/1        192.168.1.66    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to
up
```

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d) PC3 - 192.168.1.14
e) PC4 - 192.168.1.15

use subnet mask 255.255.255.224 and default gateway 192.168.1.1

10. Configure the IP addresses of PC5 - PC9 in the 2nd subnet.

a) PC5 - 192.168.1.41
b) PC6 - 192.168.1.42
c) PC7 - 192.168.1.43
d) PC8 - 192.168.1.44
e) PC9 - 192.168.1.45

Use subnet mask 255.255.255.224 and default gateway 192.168.1.33.

13. Configure static routes on R0 and R1 so that packets can travel between the 2 LAN subnets

14. Verify the router interfaces using the command : "show ip interface brief".

15. Test connectivity between PCs using the Ping command from the command prompt.

16. Ping the default gateway from each PC and then Ping PC's belonging to the other subnet.

17. If successful replies are received with no packet loss, the subnetted network is functioning correctly.

Result :-

Thus, the subnetting was implemented successfully in CISCO Packet Tracer simulator.